

Electromagnetism & Electrodynamics HW8

Daniel Haim Breger, 316136944

Technion

(Date: July 13, 2026)

1 Shell with Alternating Potential

1.1 Potential in the shell

The solution for the laplace equation in spherical coordinates is

$$\phi(r, \theta, \varphi) = \sum_{l=0}^{\infty} \sum_{m=-l}^l \left(A_{l,m} \left(\frac{r}{a} \right)^l + B_{l,m} \left(\frac{a}{r} \right)^{-(l+1)} \right) Y_{l,m}(\theta, \varphi) \quad (1)$$

But since we know the potential is not infinite at the origin, $B_{l,m} = 0$. Therefore

$$\phi(r, \theta, \varphi) = \sum_{l=0}^{\infty} \sum_{m=-l}^l A_{l,m} \left(\frac{r}{a} \right)^l Y_{l,m}(\theta, \varphi) \quad (2)$$

For convenience we'll define

$$\Phi = \begin{cases} V & \frac{2\pi k}{n} < \varphi < \frac{(2k+1)\pi}{n} \\ -V & \frac{(2k+1)\pi}{n} < \varphi < \frac{2(k+1)\pi}{n} \end{cases} \quad (3)$$

Next, we know the potential at $r = a$:

$$\phi(a, \theta, \varphi) = \sum_{l=0}^{\infty} \sum_{m=-l}^l A_{l,m} Y_{l,m}(\theta, \varphi) = \Phi \quad (4)$$

And since the spherical harmonics are a complete orthonormal set, the coefficients thus must be

$$A_{l,m} = \int_0^{2\pi} d\varphi \int_0^{\pi} d\theta \sin \theta Y_{l,m}^*(\theta, \varphi) \Phi \quad (5)$$

1.2 Zero Coefficients

We know that the following relation holds for Legendre polynomials:

$$P_{l,m}(-x) = (-1)^{l+m} P_{l,m}(x) \quad (6)$$

and we know that

$$Y_{l,m} \propto P_{l,m} \quad (7)$$

Finally we know that the problem is symmetric for rotations of π radians in the θ direction. This is important because $-\cos \theta = \cos(\pi - \theta)$ Therefore we find that

$$Y_{l,m}(\theta, \varphi) = (-1)^{l+m} Y_{l,m}(\pi - \theta, \varphi) \quad (8)$$

Which means that we must have $l + m$ to be even.

1.3 Symmetry

From the question's conditions we find that

$$V(\varphi + \Delta\varphi) = -V(\varphi) \quad (9)$$

We also note that the laplacian operator is invariant for rotational translations, therefore we must find that

$$\phi(\varphi + \Delta\varphi) = -\phi(\varphi) \tag{10}$$

i.e we find that $A = -1$.

1.4 When it is zero

Since we found the symmetry for the problem, the solution we found earlier must also satisfy it, therefore the coefficients for each $Y_{l,m}$ must satisfy it.

$$e^{im\varphi} = -e^{im(\varphi + \frac{\pi}{n})} \tag{11}$$

therefore

$$m = (2k + 1)n \tag{12}$$

so we only have the odd coefficients surviving. Combining this with the requirements that $m \leq l$ and $m + l$ is even, setting the minimal $k = 0$ we find that we must have $l = n$.

2 Three Configurations

Before starting I notice that the azimuthal angle, which is usually called φ , is θ here. I am unhappy about this, but it's merely a name change. First we'll find the volumetric charge distribution. All the charges are the same distance from the origin and on the same plane and alternate in sign, and are equally spaced along the φ axis. Therefore

$$\rho(r, \theta, \varphi) = \frac{\delta(r - R)\delta(\theta - \frac{\pi}{2})}{R^2 \sin \varphi} \sum_{n=0}^{N-1} (-1)^n \delta\left(\varphi - \frac{2\pi n}{N}\right) \quad (13)$$

As we know we can develop the potential using spherical harmonics:

$$\phi(\mathbf{x}) = \sum_{l=0}^{\infty} \sum_{m=-l}^l \frac{4\pi q_{l,m}}{(2l+1)r^{l+1}} Y_{l,m}(\theta, \varphi) \quad (14)$$

where

$$q_{l,m} = \iiint_V d\mathbf{x} Y_{l,m}^*(\mathbf{x}) |\mathbf{x}|^l \rho(\mathbf{x}) \quad (15)$$

we'll solve this in spherical coordinates and using the density from equation 13:

$$q_{l,m} = \int_0^{2\pi} d\theta \int_0^{\pi} d\varphi \sin \varphi \int_0^{\infty} dr r^2 Y_{l,m}^*(\theta, \varphi) r^l \rho(r, \theta, \varphi) \quad (16)$$

$$= \int_0^{2\pi} d\theta \int_0^{\pi} d\varphi \sin \varphi \int_0^{\infty} dr r^{l+2} Y_{l,m}^*(\theta, \varphi) \left(\frac{\delta(r - R)\delta(\theta - \frac{\pi}{2})}{R^2 \sin \varphi} \sum_{n=0}^{N-1} (-1)^n \delta\left(\varphi - \frac{2\pi n}{N}\right) \right) \quad (17)$$

$$= \frac{1}{R^2} \sum_{n=0}^{N-1} (-1)^n \int_0^{2\pi} d\theta \delta(\theta - \frac{\pi}{2}) \int_0^{\pi} d\varphi \delta(\varphi - \frac{2\pi n}{N}) Y_{l,m}^*(\theta, \varphi) \int_0^{\infty} dr r^{l+2} \delta(r - R) \quad (18)$$

$$= R^l \sum_{n=0}^{N-1} (-1)^n Y_{l,m}^*\left(\varphi = \frac{\pi}{2}, \theta = \frac{2\pi n}{N}\right) \quad (19)$$

Finally before moving to each case we notice that the total charge in all cases is 0, therefore $Y_{0,0} = \text{const}$ in all of these cases so we ignore it.

2.1 2 charges

In this case we have $N = 2$ therefore

$$q_{l,m} = R^l \sum_{n=0}^1 (-1)^n Y_{l,m}^*\left(\varphi = \frac{\pi}{2}, \theta = \pi n\right) \quad (20)$$

and

$$Y_{1,-1} = \frac{1}{2} \sqrt{\frac{3}{2\pi}} \sin \varphi e^{-i\theta} \quad Y_{1,0} = 0 \quad Y_{1,1} = -\frac{1}{2} \sqrt{\frac{3}{2\pi}} \sin \varphi e^{i\theta} \quad (21)$$

therefore

$$q_{1,\pm 1} = 2R \cdot \mp \frac{1}{2} \sqrt{\frac{3}{2\pi}} e^{\pm i\theta} = \mp R \sqrt{\frac{3}{2\pi}} \quad (22)$$

and thus the potential is

$$\phi(r, \theta, \varphi) = \sum_{m=-1}^1 \frac{4\pi q_{1,m}}{3r^2} Y_{1,m} \quad (23)$$

$$= \frac{4\pi R}{3r^2} \sqrt{\frac{3}{2\pi}} \cdot \frac{1}{2} \sqrt{\frac{3}{2\pi}} \sin \varphi (e^{i\theta} + e^{-i\theta}) \quad (24)$$

$$= \frac{2R}{r^2} \sin \varphi \cos \theta \quad (25)$$

2.2 4 charges

In this case we have $N = 4$ therefore

$$q_{l,m} = R^l \sum_{n=0}^3 (-1)^n Y_{l,m}^*(\varphi = \frac{\pi}{2}, \theta = \frac{\pi n}{2}) \quad (26)$$

From the symmetry of the problem, specifically that the charges are distributed like a plus shape, and general familiarity with the spherical harmonics, we can tell that $q_{1,l} = 0$ because they are anti-symmetric when observing their behavior in rotations of $\frac{\pi}{2}$. This can be easily seen in any of the online animations of spherical harmonics. When 1 quarter becomes positive, the adjacent ones are negative. In conclusion, we begin our search at $l = 2$:

$$Y_{2,-2} = \frac{1}{4} \sqrt{\frac{15}{2\pi}} \sin^2 \varphi e^{-2i\theta} \quad Y_{2,-1} = \frac{1}{2} \sqrt{\frac{15}{2\pi}} \sin \varphi \cos \varphi e^{-i\theta} \quad (27)$$

$$Y_{2,0} = \frac{1}{4} \sqrt{\frac{5}{\pi}} (3 \cos^2 \varphi - 1) \quad (28)$$

$$Y_{2,1} = -\frac{1}{2} \sqrt{\frac{15}{2\pi}} \sin \varphi \cos \varphi e^{i\theta} \quad Y_{2,2} = \frac{1}{4} \sqrt{\frac{15}{2\pi}} \sin^2 \varphi e^{2i\theta} \quad (29)$$

therefore

$$q_{2,0} = R^2 \sum_{n=0}^3 (-1)^n Y_{2,0}^*(\varphi = \frac{\pi}{2}, \theta = \frac{\pi n}{2}) \quad (30)$$

$$= R^2 [Y_{2,0}^*(\pi/2, 0) - Y_{2,0}^*(\pi/2, \pi/2) + Y_{2,0}^*(\pi/2, \pi) - Y_{2,0}^*(\pi/2, 3\pi/2)] \quad (31)$$

but we notice that $Y_{2,0}(\theta, \varphi + \pi) = -Y_{2,0}(\theta, \varphi)$ therefore

$$q_{2,0} = 0 \quad (32)$$

next, for $q_{2,\pm 1}$, we notice the exact same, therefore

$$q_{2,\pm 1} = 0 \quad (33)$$

Finally

$$q_{2,2} = R^2 \sum_{n=0}^3 (-1)^n Y_{2,2}^*(\varphi = \frac{\pi}{2}, \theta = \frac{\pi n}{2}) \quad (34)$$

$$= R^2 \left[Y_{2,2}^*(\frac{\pi}{2}, 0) - Y_{2,2}^*(\frac{\pi}{2}, \frac{\pi}{2}) + Y_{2,2}^*(\frac{\pi}{2}, \pi) - Y_{2,2}^*(\frac{\pi}{2}, \frac{3\pi}{2}) \right] \quad (35)$$

$$= R^2 \frac{1}{4} \sqrt{\frac{15}{2\pi}} \left[\sin^2 \frac{\pi}{2} - \sin^2 \frac{\pi}{2} e^{-i\pi} + \sin^2 \frac{\pi}{2} e^{-2i\pi} - \sin^2 \frac{\pi}{2} e^{-3i\pi} \right] \quad (36)$$

$$= \frac{R^2}{4} \sqrt{\frac{15}{2\pi}} [1 + 1 + 1 + 1] \quad (37)$$

$$= R^2 \sqrt{\frac{15}{2\pi}} \quad (38)$$

and from the rules of the multipoles we know that $q_{2,-2} = -R^2 \sqrt{\frac{15}{2\pi}}$ thus

$$\phi(x) = \frac{4\pi}{5r^3} R^2 \sqrt{\frac{15}{2\pi}} (Y_{2,2} + Y_{2,-2}) = \frac{3R^2}{5r^3} \sin^2 \varphi \cos(2\theta) \quad (39)$$

2.3 6 charges

Using the same logic as N=4, we find that all moments with $l = 0, 1, 2$ are all zero, as well as $q_{3,m}$ for all $m \in \{0, 1, -1, 2, -2\}$. Therefore

$$q_{3,3} = 6R^3 \left(\frac{1}{8} \sqrt{\frac{35}{\pi}} \right) = \frac{3R^3}{4} \sqrt{\frac{35}{\pi}} \quad (40)$$

Therefore

$$\phi(x) = \frac{4\pi}{7r^4} \frac{3R^3}{4} \sqrt{\frac{35}{\pi}} (-Y_{3,3} + Y_{3,-3}) \quad (41)$$

$$= \frac{3\pi R^3}{7r^4} \sqrt{\frac{35}{\pi}} \cdot \frac{1}{8} \sqrt{\frac{35}{\pi}} \sin^3 \varphi \cos(3\theta) \quad (42)$$

$$= \frac{15R^3}{4r^4} \sin^3 \varphi \cos(3\theta) \quad (43)$$

3 Six Charges

3.1 zero monopole dipole quadropole

We'll show this from symmetry considerations.

3.1.1 Monopole

The total charge is 0 therefore the monopole is zero.

3.1.2 Dipole

If we look at rotations of 60 degrees, the rotated charge distribution is exactly the negative of the original. On the other hand, the dipole is a vector. This means we have

$$\mathbf{p} \rightarrow -\mathbf{p} \quad (44)$$

after a rotation of 60 degrees. This is only possible if $\mathbf{p} = 0$.

3.1.3 Quadropole

the quadropole moments are

$$q_{2j} = \int d\mathbf{r} Y_{2j}^*(\mathbf{r}) r^2 \rho(\mathbf{r}) \quad (45)$$

but

$$Y_{2j}^* \rightarrow (-1)^2 Y_{2j}^*(-r) = Y_{2j}^*(-r) \quad (46)$$

so the quadropole should remain identical after a mirroring, but we immediately see the charge distribution would become its own negative, so the quadropole moment must also be zero.

3.2 3rd multipole

$$q_{33} = \sum_{\alpha} q_{\alpha} r_{\alpha}^3 Y_{33}^*(\theta_{\alpha}, \varphi_{\alpha}) \quad (47)$$

$$= \sum_{\alpha} q_{\alpha} r_{\alpha}^3 \left(-\frac{1}{4} \sqrt{\frac{35}{\pi}} \sin^3 \theta e^{3i\varphi} \right) \quad (48)$$

but $r_{\alpha} = a$ and $\theta_{\alpha} = \pi/2$ and $q_{\alpha} = (-1)^{\alpha+1} q$ and $\varphi_{\alpha} = \frac{\pi}{6} (2\alpha+1)$ so

$$q_{33} = -\sqrt{\frac{35}{16\pi}} a^3 q \sum_{\alpha} (-1)^{\alpha+1} e^{-\frac{i\pi}{2}(2\alpha+1)} \quad (49)$$

$$= -6 \sqrt{\frac{35}{16\pi}} a^3 q \quad (50)$$

3.3 power of N

since the potential depends on the radius $\phi \propto r^{-(l+1)}$ but we saw that the first non-zero multiple is for $l = 3$ it means that $N = 4$.